

## SECTION B

7 (**short**) A metal on heating undergoes a change from a body-centred cubic (BCC) structure to a face-centred cubic (FCC) structure. At the change temperature, the lattice constant  $a$  has a value of 0.28 nm for the BCC structure and 0.35 nm for the FCC structure.

(a) Sketch the unit cells of the BCC and FCC structures, showing the atomic sites. Determine the atomic density of the metal in atoms per  $\text{m}^3$  for each structure at the change temperature. [6]

(b) Estimate the fractional change in length  $\Delta L / L$  of a bar of the metal as it is heated through the change temperature. It may be useful to note that

$$\frac{\Delta L}{L} \approx \frac{1}{3} \frac{\Delta V}{V}$$

where  $\Delta V / V$  is the fractional change in volume. [4]

## 8 (**short**)

(a) Define the terms *hardness* and *yield stress*. A metal deforms in a standard tensile test according to the following relationship between true stress  $\sigma_t$  and true strain  $\epsilon_t$ :

$$\sigma_t = 450\sqrt{\epsilon_t} \text{ MPa.}$$

In a forming operation a cylinder of this material is stretched in simple tension such that its length increases by 25%. Estimate the yield stress and Vickers hardness of the metal after forming. [6]

(b) An identical cylinder of the same metal is deformed in simple compression. What percentage change in cylinder length will produce the same hardness as that found in the stretched cylinder? Find the ratio of the loads at the end of the two forming operations. [4]